

JAVA PROGRAMMING

Chapter 2: Variables and Literals

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1. Java Variables

A variable is a named memory location used to store data. Think of it as a labeled box that holds values your program needs to work with.

Every variable in Java must have:

- A data type (int, double, String, etc.)
- A unique name
- A value (optional at declaration)

2. Creating Variables in Java

Let's create a variable that stores the number of problems solved:

```
int problemsSolved = 120;
```

int	→ data type
problemsSolved	→ variable name
120	→ value stored

This means we are storing the value 120 inside a memory location named **problemsSolved**.

3. Declaration and Assignment

You can declare and assign separately:

```
double completionRate;  
completionRate = 87.5;
```

Or do both together:

```
boolean isPremiumMember = true;
```

Both approaches are correct.

4. Java is Statically Typed

Java requires that every variable be declared before it is used. The compiler needs to know the data type.

✓ This is valid:

```
int totalContests = 15;
```

■ This is invalid:

```
totalContests = 15; // Variable not declared
```

Java must know the data type first.

5. Changing Variable Values

The value stored inside a variable can change. The memory location stays the same, but the value updates.

```
int rank = 450;  
// after next contest  
rank = 320;
```

That's why it is called a variable — because the value can vary over time.

6. Important Rule: Type Cannot Change

Once declared, a variable cannot change its data type. Java does not allow redeclaration with a different type in the same scope.

■ This is invalid:

```
int streak = 30;  
double streak = 30.5;    // Not allowed
```

Once streak is declared as int, it will always be int. You cannot reassign it as double.

7. Naming Rules for Variables

Follow these rules:

- Can contain letters, digits, underscore (`_`), and dollar sign (`$`)
- Cannot start with a digit
- Cannot use Java keywords (like `int`, `class`, `public`, etc.)
- Case-sensitive (`score` and `Score` are different variables)

Example of Case-Sensitivity:

```
int score;  
int Score; // different variable
```

8. Java Literals

A literal is the actual fixed value assigned to a variable.

```
int activeUsers = 5000;
```

activeUsers	→ variable
5000	→ literal

9. Integer Literals

Integer literals represent whole numbers. Java supports 4 different formats:

1■■■ Decimal (Base 10)

```
int enrolledStudents = 250;
```

2■■■ Binary (Base 2)

```
int accessMask = 0b10101; // 0b indicates binary
```

3■■■ Octal (Base 8)

```
int filePermission = 075; // Leading 0 indicates octal
```

4■■■ Hexadecimal (Base 16)

```
int themeColor = 0xFF; // 0x indicates hexadecimal
```

Integer literals are used with: byte, short, int, long

10. Floating-Point Literals

Floating-point literals represent decimal numbers.

```
double averageScore = 78.6;  
float discountPercent = 15.5F;
```

Important: By default, decimal numbers are treated as double. To store in float, you must use F suffix.

Scientific Notation:

```
double serverLoad = 2.5e3;  
// This means:  $2.5 \times 10^3 = 2500$ 
```

This format is useful in data analytics and performance calculations.

11. Character Literals

Character literals represent a single character enclosed in single quotes.

```
char grade = 'A';
```

You can also use escape sequences:

```
char newLine = '\n';    // new line  
char tabSpace = '\t';    // tab space
```

Escape characters represent special formatting like newlines, tabs, and quotes.

12. String Literals

A string literal is a sequence of characters inside double quotes.

```
String platformName = "CodeHelp ONE";  
String courseTrack = "DSA + Core + Aptitude";
```

Anything inside double quotes is treated as text. Unlike character literals which use single quotes, strings can contain multiple characters.

13. Boolean Literals

Boolean literals represent logical values. They can only be true or false.

```
boolean isLoggedIn = true;  
boolean hasCompletedSheet = false;
```

Only true and false are valid boolean literals. They are used in conditional statements and loops.

14. Why Variables Matter for Placements

In interviews, companies test:

- Variable declaration clarity
- Correct data type usage
- Memory understanding
- Type mismatch errors
- Edge case handling

Most beginner mistakes happen because of poor understanding of:

- Data types and type casting
- Literal formats (decimal, binary, hex)
- Implicit type assumptions

■ Key Insight:

Strong fundamentals here prevent errors later in DSA and system-level programming. Interviewers often ask tricky questions about type conversions and memory allocation.

15. Quick Summary & Practice

Concept	Description
Variable	Named memory location to store data
Data Type	Determines what kind of value can be stored
Literal	The actual fixed value in the code
Statically Typed	Type must be declared before use
Type Immutable	Variable type cannot change after declaration

—■ Practice Task:

Try creating variables for:

- Your contest rating
- Total problems solved
- Platform subscription status (true/false)
- Your name

Run them in your IDE and observe the behavior. Try to use different literal formats (binary, hex) for numbers.

Remember: Consistent practice builds clarity. Start with simple variables and gradually explore different data types and formats.